

ANNETTE ISLAND, AK, USA

WMO: 703980

Lat: 55.039N

Lon: 131.579W

Elev: 33

StdP: 100.93

Time Zone: -9.00 (NAK)

Period: 96-19

WBAN: 25308

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-7.6	-5.4	-18.6	0.7	-4.7	-15.6	1.0	-2.9	14.2	6.8	12.6	6.8	2.6	60	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	23.9	16.6	21.6	15.8	19.6	14.7	17.5	22.7	16.4	20.5	15.4	18.5	3.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	10.9	18.4	14.6	10.4	17.4	14.0	10.0	16.6	49.1	22.8	45.7	20.5	43.1	18.5	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.2	8.2	-10.9	28.2	2.7	2.4	-12.8	29.9	-14.4	31.3	-16.0	32.7	-18.0	34.5		
			-11.6	19.1	2.7	1.2	-13.5	19.9	-15.1	20.6	-16.6	21.2	-18.5	22.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	3.0	3.0	4.0	6.8	10.3	13.1	14.9	15.1	12.3	8.5	4.8	2.8
	DBStd	5.40	4.09	3.13	2.85	2.36	2.77	2.54	2.34	2.23	2.01	2.42	3.19	3.63
	HDD10.0	1174	216	197	187	102	29	1	0	0	4	57	157	224
	HDD18.3	3698	474	431	445	347	248	161	113	104	182	304	406	482
	CDD10.0	530	0	0	0	5	39	93	151	159	72	11	1	0
	CDD18.3	12	0	0	0	0	0	3	5	4	0	0	0	0
	CDH23.3	99	0	0	0	0	3	32	38	25	1	0	0	0
	CDH26.7	17	0	0	0	0	0	8	7	2	0	0	0	0
Wind	WSAvg	3.3	3.8	3.4	3.6	3.4	2.9	2.8	2.6	2.6	2.9	3.5	3.6	3.8
Precipitation	PrecAvg	2643	264	214	209	192	149	123	121	173	251	367	298	281
	PrecMax	3662	489	459	599	429	373	238	230	373	461	690	632	592
	PrecMin	1973	76	3	54	49	13	16	14	20	64	117	92	87
	PrecStd	343	90	103	96	85	77	52	56	85	94	122	112	102
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.4	13.6	18.5	23.1	27.6	27.7	26.5	21.3	15.9	13.3	11.1
		MCWB	6.2	5.7	8.0	9.8	13.4	17.0	19.1	18.1	15.7	11.3	8.3	6.3
	2%	DB	9.6	9.6	10.4	14.7	19.8	22.5	23.6	23.2	18.4	14.1	11.2	9.7
		MCWB	7.5	5.9	6.5	8.3	12.2	15.6	17.3	17.1	14.1	10.5	8.6	7.2
	5%	DB	8.6	8.3	9.0	12.2	17.1	19.5	20.8	21.0	16.7	12.8	10.0	8.6
		MCWB	7.0	5.7	6.3	7.6	11.5	13.9	15.9	16.0	13.3	10.3	8.2	6.9
	10%	DB	7.8	7.3	7.9	10.4	14.9	17.1	18.7	19.1	15.3	11.9	9.0	7.7
		MCWB	6.4	5.6	5.7	6.9	10.5	13.1	14.9	15.2	12.7	10.0	7.5	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	8.0	8.9	11.4	15.0	17.8	19.4	18.7	16.4	13.4	10.5	9.0
		MCDB	9.5	9.1	10.9	16.6	21.0	25.3	26.5	24.9	20.6	14.8	11.2	9.9
	2%	WB	8.0	7.0	7.8	9.4	13.1	15.8	17.7	17.4	15.0	12.1	9.4	8.0
		MCDB	8.7	7.9	9.6	12.8	17.7	21.2	23.0	22.3	17.3	12.7	10.5	8.9
	5%	WB	7.5	6.5	6.9	8.3	12.0	14.7	16.4	16.5	14.1	11.2	8.5	7.3
		MCDB	8.3	7.6	8.4	11.1	15.5	18.7	20.1	20.3	15.7	11.9	9.7	8.3
	10%	WB	6.7	5.7	6.2	7.6	11.0	13.6	15.2	15.6	13.4	10.5	7.8	6.4
		MCDB	7.6	7.1	7.6	9.9	13.9	16.5	18.0	18.4	14.7	11.4	8.8	7.4
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.3	5.4	6.0	6.4	6.0	5.7	5.9	5.5	5.0	4.8	4.5
		MCDBR	4.5	5.9	6.9	9.3	10.9	11.3	10.4	9.9	8.2	6.3	5.1	4.3
		MCWBR	3.2	3.8	4.2	4.7	5.3	5.1	5.0	4.8	4.7	4.0	3.8	3.2
	5% WB	MCDBR	3.5	4.5	5.8	7.5	9.2	9.7	9.7	9.1	6.9	4.4	4.3	3.8
		MCWBR	3.2	3.6	4.1	4.3	4.9	4.7	4.7	4.6	4.3	3.6	3.7	3.4
Clear-Sky Solar Irradiance	taub	0.300	0.324	0.334	0.366	0.363	0.367	0.373	0.359	0.345	0.335	0.308	0.293	
	taud	2.392	2.413	2.450	2.367	2.442	2.462	2.479	2.543	2.557	2.478	2.356	2.292	
	Ebn at Noon	647	762	844	858	882	880	867	860	826	739	618	550	
	Edh at Noon	51	71	85	106	104	104	100	89	77	66	52	45	
All-Sky Solar Radiation	RadAvg	0.48	1.27	2.25	3.70	4.67	4.88	4.36	3.81	2.48	1.36	0.61	0.38	
	RadStd	0.06	0.25	0.36	0.28	0.47	0.38	0.55	0.37	0.32	0.21	0.10	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page